

$$\begin{aligned}
 \text{the likelihood: } L(\pi) &= \prod_{i=1}^N f(y_i) \\
 &= \prod_{i=1}^N \pi^{y_i} (1 - \pi)^{1-y_i} \\
 &= (\textit{could simplify as before if we wanted...})
 \end{aligned}$$